Nicholas Penzel

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PROFESSIONAL SUMMARY

Emerging Professional and graduate student in Ecosystem Science and Sustainability with a specialized focus on water resources and environmental data analysis. Expertise in water quality analysis, chemistry, hydrologic modeling, watershed assessment, and ecosystem monitoring. Skilled in leveraging statistical computing and geospatial analysis to address complex water resource challenges. Background in conservation biology and ecology, with a strong commitment to sustainable water management. My core competencies include:

- Data analysis in RStudio and Excel, GIS analysis in R, ArcGIS, and Google Earth Engine
- Team and project management
- Microsoft Office Suite
- Technical and scientific writing and document production
- Colorado water law and governance
- Hydrologic modeling and analysis in HEC-HMS and RStudio
- Data entry, database management, and QAQC

RESEARCH EXPERIENCE

Colorado Division 5 ET & Consumptive Use Dashboard **2025**

Colorado State University, Fort Collins, CO

- Developed a web application to analyze agricultural consumptive water use in Colorado Water Division 5, providing insights for water resource management
- Integrated state and national datasets to model consumptive use of irrigation infrastructure relative to diverted water volumes, enabling data-driven decision-making for agricultural water usage

Hydrology Coursework & Field Research **2025**

Colorado State University, Fort Collins, CO

- Wrangled and analyzed large hydrologic datasets incorporating multiple data sources (USGS, NOAA, PRISM, state water agencies) for integrated watershed and water quality analysis
- Developed expertise in Colorado water law, interstate compacts, and water administration
- Conducted comprehensive watershed assessments, including water balance modeling, environmental analysis, and hydrologic characterization across multiple Colorado watersheds
- Performed stream discharge measurements using handheld acoustic Doppler velocimeters (ADV), salt tracer dilution tests, and USGS gauge analysis

Seasonal Research Technician, Northern Rocky Mountain Science Center **2021-2022**

U.S. Geological Survey, Bozeman, MT

- Collected and analyzed ecological data to model American pika population dynamics in response to climate change and habitat fragmentation across montane ecosystems
- Maintained and contributed to project SOPs for dataset digitization and management
- Performed QAQC on project databases to ensure downstream analysis accuracy
- Conducted independent field surveys in remote locations across the Western U.S., including 4-10 day backcountry expeditions requiring advanced navigation and field skills
- Collaborated with federal land management agencies (National Park Service, BLM) and private landowners to facilitate cooperative research and data sharing
- Co-authored peer-reviewed publications on climate adaptation and species conservation

Spruce Bark Beetle Remote Sensing Research **2021**

Colorado College, Colorado Springs, CO

- Developed a methodology to detect Engelmann spruce mortality across broad spatial scales using Google Earth Engine and a random forest classifier
- Allowed for rapid, broad analysis of stand mortality and post hoc analysis of potential causes for epidemic spruce beetle outbreaks

MANAGEMENT EXPERIENCE

Store Manager **2021-2025**

Cripple Creek Bike & Backcountry, Carbondale, CO

- Managed a multi-million dollar retail business, demonstrating strong organization and leadership
- Led project planning, timeline management, day-to-day operations, and team coordination
- Led SEO and content generation projects, including blogs, video, photography, and social media
- Developed SOPs and implemented them company-wide to increase efficiency and customer satisfaction

COMMUNICATIONS EXPERIENCE

Reporter, Pueblo Pulp, Pueblo, CO **2020**

- Paid intern covering news and culture in Southern Colorado, focusing on underrepresented communities and environmentalism topics such as outdoor recreation, COVID response, and Black Lives Matter protests
- Photographed and wrote contextual coverage on a broad range of topics
- Concisely communicate to a wide audience on complicated issues

EDUCATION

Professional Science Master's Degree in Ecosystem Science and Sustainability **Dec. 2026**

Colorado State University, Fort Collins, CO

Bachelor of Arts in Organismal Biology and Ecology, Cum Laude **May 2021**

Colorado College, Colorado Springs, CO **GPA: 3.88**

Study Abroad: African Ecology and Conservation **2020**

Organization for Tropical Studies, Skukuza, South Africa

PUBLICATIONS

Geographic and taxonomic variation in adaptive capacity among mountain-dwelling small mammals: implications for conservation status and actions. (2023) E.A. Beever, J.L. Wilkening, P.D. Billman, et al. *Biological Conservation, 282*

Evaluating strength of evidence and drivers for species-climate relationships across space and time. E.A Beever, M. Peacock, J. Wilkening, et al. *Conservation Biology*. *In review*

HONORS AND AWARDS

Boettcher Scholarship Recipient **2017-2021**

Dean's List, Colorado College **2019-2021**

Mary Alice Hamilton Award in Organismal Biology and Ecology **2021**